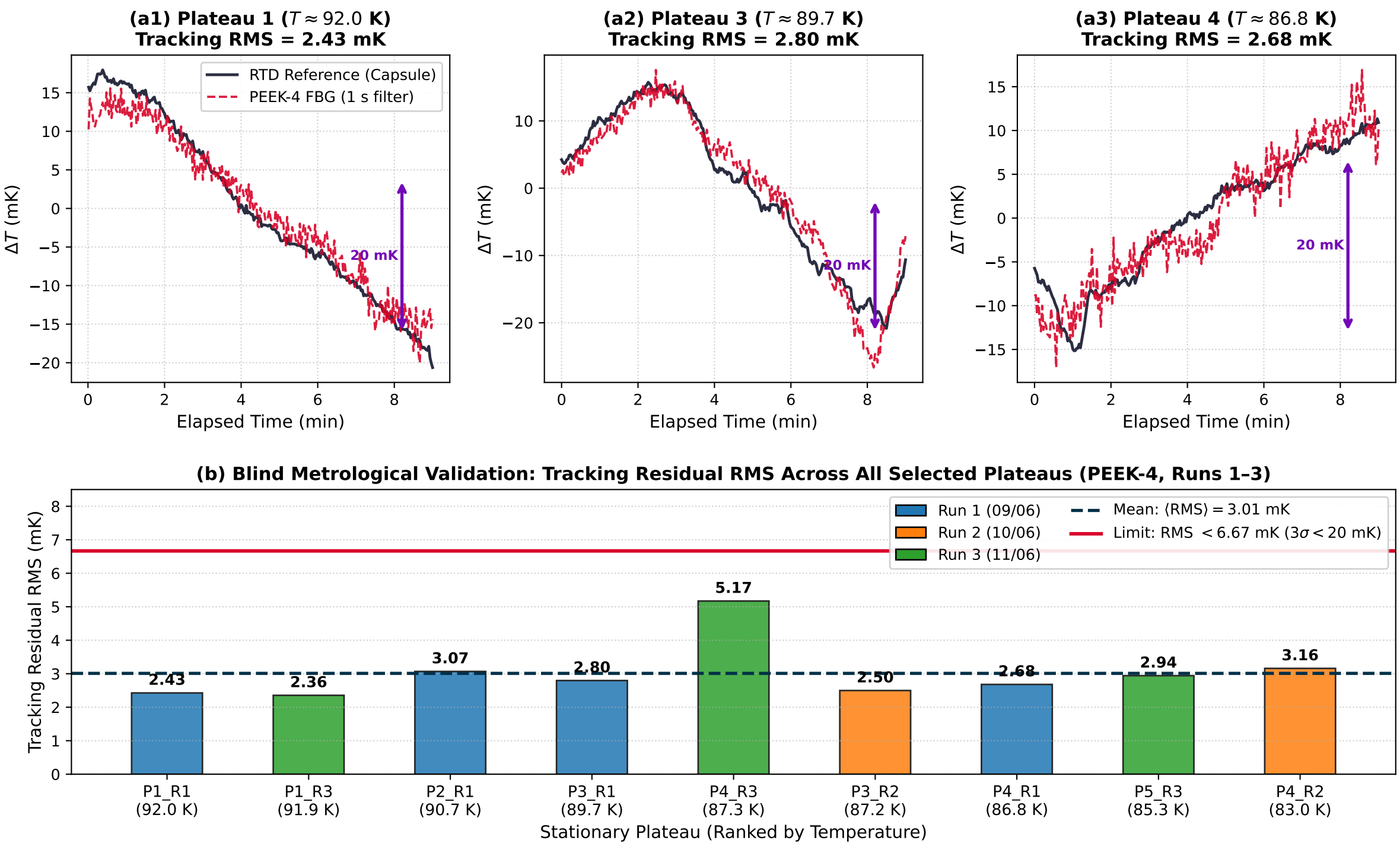
(a1) Plateau 1 ($T \approx 92.0$ K)
Tracking RMS = 2.43 mK
RTD Reference (Capsule)
PEEK-4 FBG (1 s filter)
20 mK
ΔT (mK)
Elapsed Time (min)
(a2) Plateau 3 ($T \approx 89.7$ K)
Tracking RMS = 2.80 mK
20 mK
ΔT (mK)
Elapsed Time (min)
(a3) Plateau 4 ($T \approx 86.8$ K)
Tracking RMS = 2.68 mK
20 mK
ΔT (mK)
Elapsed Time (min)
(b) Blind Metrological Validation: Tracking Residual RMS Across All Selected Plateaus (PEEK-4, Runs 1–3)
Run 1 (09/06)
Run 2 (10/06)
Run 3 (11/06)
Mean: ⟨RMS⟩ = 3.01 mK
Limit: RMS < 6.67 mK (3σ < 20 mK)
Tracking Residual RMS (mK)
2.43
2.36
3.07
2.80
5.17
2.50
2.68
2.94
3.16
P1_R1 (92.0 K)
P1_R3 (91.9 K)
P2_R1 (90.7 K)
P3_R1 (89.7 K)
P4_R3 (87.3 K)
P3_R2 (87.2 K)
P4_R1 (86.8 K)
P5_R3 (85.3 K)
P4_R2 (83.0 K)
Stationary Plateau (Ranked by Temperature)